



CHARACTERIZATION OF 17 NOVEL MICROSATELLITE LOCI FOR THE BROWN SPINY LOBSTER *PANULIRUS ECHINATUS* SMITH, 1869 USING MISEQ

BY

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ABSTRACT

Using high-throughput sequencing technology 17 polymorphic microsatellites with perfect tetra-nucleotide repeats were identified for the brown spiny lobster *Panulirus echinatus* Smith, 1869. Two to ten alleles were detected per locus across 30 samples analysed from the Canary Archipelago. Observed and expected heterozygosities per locus ranged between 0.100 and 0.867 and 0.095 and 0.799, respectively. No significant linkage disequilibrium was found between pairs of loci, and only one locus (Pe-L43) deviated from Hardy-Weinberg equilibrium, likely due to the presence of null alleles. Despite its economic importance and the threat of overfishing, population genetics studies of this species are lacking. Therefore, these 17 novel microsatellites markers will be a useful genetic resource for future conservation studies of *P. echinatus*.

Key words. — Population genetics, endangered species, connectivity, conservation

RESUMEN

Se identificaron diecisiete microsatélites polimórficos con repeticiones de tetra-nucleótidos perfectos para la langosta herreña *Panulirus echinatus* Smith, 1869, usando tecnología de secuenciación masiva. Se detectaron de dos a diez alelos por locus a través del análisis de 30 muestras del archipiélago canario. Los valores de heterocigosis observados y esperados por locus oscilaron entre 0,100 y 0,867 y 0,095 y 0,799, respectivamente. No se encontró un desequilibrio de ligamiento significativo entre pares de loci, y solo un locus (Pe-L43) se desvió de lo esperado según el equilibrio de Hardy-Weinberg, probablemente debido a la presencia de alelos nulos. A pesar de su importancia económica y de la amenaza de la sobrepesca, no se cuenta con estudios de genética de poblaciones de esta especie. Por lo tanto, estos 17 nuevos marcadores microsatélites podrían ser de utilidad en futuras investigaciones para la conservación de *P. echinatus*.

Palabras clave. — Genética de poblaciones, especies amenazadas, conectividad, conservación

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INTRODUCTION

The brown spiny lobster *Panulirus echinatus* Smith, 1869 (Decapoda: Palinuridae) is a commercially valuable fishery resource, particularly for its use in the food industry. Harvesting of this lobster species has greatly increased in the last decade (Pinheiro & Lins-Oliveira, 2006; Freitas et al., 2007; Moro et al., 2014), leading to its overexploitation and a reduction in the number and size of populations. As a result, this species has been classified as endangered in certain localities, such as in the Canary Islands (Moro et al., 2014), despite having a wide distribution in the Atlantic, particularly in the Central Atlantic Islands (Canary, Cape Verde, Ascension and St. Helena islands) and along the northeastern coast of Brazil (Maranhão to Rio de Janeiro, including the São Pedro and São Paulo Archipelago, Fernando de Noronha, Atol das Rocas, the Abrolhos Archipelago and the Trindade Islands) (Manning & Chace, 1990; Coelho & Ramos-Porto, 1998; Melo, 1999; Coelho et al., 2007; Faria-Júnior et al., 2013; Gaeta et al., 2015). Species of the genus *Panulirus* are characterized by the presence of a phyllosoma, a larval phase adapted for long-term dispersal (Palero et al., 2008). The dispersal ability of this group is among the greatest found for crustaceans as phyllosoma larvae can live in the plankton from several months to two years and can be transported over long distances by ocean currents (Sarver et al., 2000). Given the long larval period and long distance dispersal abilities of these lobsters, metapopulations are very common. Despite this, the wide and discontinuous distribution of this species may pose a problem for its long-term survival due to potential interruptions or reductions in gene flow. Furthermore, for some populations factors, such as currents, temperatures and habitat, also directly influence the recruitment and settlement of lobsters in bordering areas.

The Canary Island populations of *P. echinatus* have drastically declined over the last couple of decades, and current population sizes are small (Riera et al., 2014). Indeed, in some localities, such as in shallow caves of the western Canary Islands, sightings of this species are rare (Moro et al., 2014), despite being frequently found there in the past (González-Pérez, 1995; Templado et al., 2004). Even at larger geographic scales, for example, on the island of La Gomera, its presence is rare (Anonymous, 2005). As a result, this species is currently categorized as endangered in the Canarian Catalogue (Royal Decree 139/2011, of 4 February 2011 for the development of the list of wild species under special protection regime and the Spanish Catalogue of Threatened Species). To date, little to no information is known about the population dynamics or genetic structure of *P. echinatus* in the Canary Islands.

Genetic studies can provide crucial information about the potential resilience and sustainability of species and influence criteria for conservation management

plans. Previous studies have demonstrated the utility of microsatellites as genetic markers for stock identification and management of several marine species (e.g., Was et al., 2010; Milano et al., 2011; Cuveliers et al., 2012; Corander et al., 2013; Delghandi et al., 2015; López-Márquez et al., 2016). Microsatellite markers have been developed for several *Panulirus* species (Ben-Horin et al., 2009; Kennington et al., 2010; Dao et al., 2013; Liu et al., 2013; Truelove et al., 2016), although not for *P. echinatus*. Although cross-species amplifications for *Panulirus argus* (Latreille, 1804) (cf. Diniz et al., 2004, 2005; Tringali et al., 2008) and *Panulirus homarus* (Linnaeus, 1758) (cf. Delghandi et al., 2015) have been successful, positive amplifications for distantly-related congeners, such as *P. echinatus*, are generally low (Ptacek et al., 2001; Ben-Horin et al., 2009). Therefore, the aim of this study is to characterize new highly-specific variable markers for *P. echinatus*, with the eventual goal of evaluating genetic population structure and assessing population connectivity and potential signs of bottleneck events within *P. echinatus* populations that are experiencing declines due to exploitation and lack of fisheries regulations.

Overall, these *P. echinatus* microsatellite markers will enable researchers to determine genetic diversity, measure levels of genetic connectivity, identify subpopulations genetically and guide governments in the conservation and management of this species.

MATERIAL AND METHODS

Tissue samples were collected from 30 *Panulirus echinatus* individuals found in localities with known populations in the Canary Archipelago (28°32'N 15°43'W), including the islands of Tenerife, La Palma and El Hierro. Lobsters were collected by hand while SCUBA diving. Without killing specimens, muscular tissue was extracted from the last ambulatory leg or pereopod. Specimens of this species were identified morphologically based on the presence of an antennular plate with a pair of strong thorns, rounded whitish spots covering the body, especially the tail, and yellowish or whitish longitudinal lines along the antennules and legs (which lack bands or spots). Tissue samples were preserved in absolute ethanol and stored at 4°C until processed for molecular analyses. Genomic DNA (gDNA) was extracted from approximately 50–100 mg of tissue, using the Qiagen BioSprint 15 DNA Blood Kit (Qiagen Iberia, Madrid, Spain) according to the manufacturer's protocol, including an RNase treatment step. DNA quantity and quality were checked using a Nanodrop 1000 and gel electrophoresis (1% agarose gels), respectively. DNA aliquots (2 ng/μl) were then prepared for genotyping analyses. Species identification was then confirmed by amplifying and sequencing a fragment of the

mitochondrial 16S ribosomal RNA gene, using the primer pair 16S-ar and 16S-br (Palumbi, 1996). The BLAST algorithm was used to compare sequences (Altschul et al., 1997).

An enriched genomic library was prepared from a single *P. echinatus* specimen (AR03-Pe-MNCN) from Atol das Rocas (Brazil) for Illumina Miseq next-generation sequencing (NGS). The library was prepared at the Cornell Life Sciences Core Laboratory Center as described by Nali et al. (2014) and López-Márquez et al. (2016). In short, DNA was digested with restriction enzymes, and Illumina Y-adaptors were ligated to the fragments. Fragments were then enriched for microsatellites by hybridizing with oligonucleotide repeat probes, including the tetranucleotide probes (TTTC)₆, (GATA)₇, (TTAC)_{6,75}, (GATG)_{4,25}, (TTTG)_{5,25}, (TGTC)_{4,5} and (TGTA)₆. Enriched fragments were isolated using streptavidin-coated magnetic beads. Fragments between 300 and 600 bp were selected for sequencing and genotyping.

A total of 30 MB of data, consisting of 454,250 reads, were obtained. The fasta file from the de novo assembly was scanned for potential microsatellite loci and primer pairs with the program MSATCOMMANDER (Faircloth, 2008) and QDD version 3.1 (Megléc et al., 2014). After trimming adapter sequences, an 'all against all' BLAST (Altschul et al., 1997) search, implemented in QDD version 3.1 with microsatellite motifs masked, was performed to compare sequence similarities. For further analyses, only sequences longer than 100 bp with perfect motifs that also consisted of at least five repetitions of any 2-6 bp motif were selected. More than 110,000 microsatellite-containing sequences were identified as potential markers. Using QDD to select only sequences with sufficient flanking regions for primer design, the number of candidate sequences was reduced to 1569, which was further reduced to 775 by MSATCOMMANDER after applying Excel's TRUE and FALSE function tools. Sequences with poly-A, poly-C, poly-G and poly-T stretches (≥ 7 bp) were also excluded, leaving 673 sequences. Primers were designed using PRIMER3 v.0.4.0 (Untergasser et al., 2012) with the following criteria: primer length 20-27 bp, product size 218-405 bp, GC content 40-60% and melting temperature around 60°C. Although 673 primer pairs matching this criteria could be designed, some pairs showed amplicon similarities to known transposable elements, as assessed in RepeatMasker v4.0 (Smit et al., 2013-2015), and were therefore rejected. A total of 50 primer pairs was ultimately selected and synthesized to screen for microsatellite markers.

Primers were first validated by successful amplification of 30 individuals of *P. echinatus*. Nested PCRs were carried out in a total volume of 10 μ l with 1 $\times$ PCR Biotools Standard Reaction Buffer including 2 mM MgCl₂, 0.12 μ M forward and 0.8 μ M reverse primers, 0.8 μ M primer PaulAn (5'-TGACGACCCCATGCTACG-3') (Acevedo et al., 2009) fluorescently 5'-end labelled with 6-FAM, NED, VIC or

TABLE I
Description of the 17 microsatellite loci isolated from *Panulirus echinatus* Smith, 1869 based on a sample of 30 individuals from the Canary Islands

Locus name	Primer sequences (5' → 3')	Repeat motif	Allele size range (bp)	N _a	H _o	H _e	P-value	HWE	Accession number
Pe-L3	F: 6-FAM-GCTGGAAGACACTCGAGACA R: TTTACGATTCCATTCCGTGA	(CAGA) ₁₈	201-243	8	0.800	0.734	0.7451		MF170655
Pe-L8	F: PET-GGTCAATTGTGCAGGTCACAC R: GATTGGAACCCATGCTGACT	(GACA) ₁₀	166-190	4	0.700	0.601	0.3295		MF170656
Pe-L10	F: PET-CCTGTGGAAACGATGTGGTA R: GGGACTGCACCATAACCAGA	(GACA) ₁₀	220-232	3	0.607	0.564	0.8275		MF170657
Pe-L18	F: NED-AATTGGCTGCAGTGGTTGG R: CACCGCTATGATACGGGAAT	(GACA) ₈	130-134	2	0.267	0.278	1.0000		MF170658
Pe-L22	F: 6-FAM-GAAGCTTCAGGAGAAAGCTC R: TGGTGTCACAATCCCTCTCA	(AGAC) ₈	200-204	2	0.367	0.375	1.0000		MF170659
Pe-L23	F: PET-CGAACCAAGTATGCAGAGCA R: CCTTTAGTTGGCTGTACATCA	(GACA) ₇	251-263	3	0.310	0.399	0.0174		MF170660
Pe-L24	F: VIC-AGGTGATCCAGGCAATTGTA R: TCCTCTTGGCGTTTCAGTTTC	(CAGA) ₇	145-149	2	0.200	0.278	0.1584		MF170661
Pe-L27	F: PET-TTTTCAATATGACCGGAAACG R: CCTTCCACAAGTCTTCACC	(GACA) ₇	178-186	2	0.100	0.095	1.0000		MF170662
Pe-L35	F: PET-CCTGCTTTGTTCACCTGTT R: GCCTTTCAGTCCCACAGTA	(CATA) ₁₁	276-296	6	0.600	0.739	0.0838		MF170663
Pe-L36	F: NED-CGTACAGGGGTGGAAAATA R: CGTCATGAACAGGAGTTCCA	(CATA) ₇	114-142	3	0.367	0.305	0.6331		MF170664
Pe-L37	F: 6-FAM-AGCCAGATGGAAGGGAAT R: ACACGGGAAGCATTACTTGG	(TACA) ₇	283-295	4	0.593	0.644	0.6536		MF170665
Pe-L38	F: 6-FAM-ATGTCCTGCTGGCTGTTTCT R: AACCATTACACCCAGGAAGC	(CATA) ₇	195-218	4	0.467	0.569	0.1551		MF170666
Pe-L40	F: VIC-GGCCAGAATGGGATATCAAG R: CACCCAATATTCACCCGTTA	(CATA) ₇	138-151	4	0.833	0.703	0.2850		MF170667

TABLE I
(Continued)

Locus name	Primer sequences (5' → 3')	Repeat motif	Allele size range (bp)	N_a	H_o	H_e	P -value	HWE	Accession number
Pe-L42	F: VIC-TTAGGGTGCTGCTTGCTT	(CATA) ₉	287-323	4	0.407	0.469	0.2651		MF170668
	R: ACGGGGCAACACGAGTATAA								
Pe-L43	F: NED-AACACTGATAACACACATACTTCAGA	(CATA) ₉	218-247	10	0.533	0.799	0.0009*		MF170669
	R: CGGCTTGACAACTGCATTAC								
Pe-L45	F: NED-CGGCTTATCTCTATTCGCTTT	(CATA) ₈	194-202	3	0.500	0.598	0.0962		MF170670
	R: TGCCATATTCGAGGGAAAAC								
Pe-L48	F: VIC-TCTGCCAAATGAGTGTGCTTG	(TACA) ₁₀	220-244	7	0.867	0.771	0.1610		MF170671
	R: TTGGTAAAGAAGCACTTTCTGG								

Abbreviations: F, forward; r, Reverse; N_a , number of alleles; H_o , observed heterozygosity; H_e , expected heterozygosity; HWE, Hardy-Weinberg equilibrium. Annealing temperature = 56°C.

*Loci deviating from HWE after null allele evaluation and after correction using the FALSE Discovery Rate procedure of Benjamini and Hochberg (1995) for multiple pairwise comparisons.

PET (table I), 0.2 mM of each dNTP, 0.5 U DNA polymerase (Biotools) and 2 ng of template DNA. A 5'-end tag was included in the reverse primers to avoid stutters (Brownstein, 1996). Veriti™ Thermal Cyclers (Applied Biosystems) were used to perform PCR amplifications with the following conditions: initial denaturation at 94°C for 3 min, followed by 35 cycles at 94°C for 45 s, 56°C for 45 s and 72°C for 30 s, and a final extension at 72°C for 10 min. Fluorescently-labeled PCR products were run on an ABI PRISM 3730 DNA Sequencer (Applied Biosystems) and scored using the GeneScan-500 (LIZ) size standard. GeneMapper version 4.0 (Applied Biosystems) was used for genotyping analysis.

Based on the expected sizes of the amplification products, the electropherogram patterns and polymorphism scores, 17 polymorphic microsatellite loci were chosen for further characterization. The sequences of the selected loci were deposited in GenBank (for accession numbers, see table I). The number of alleles (N_a), observed (H_o) and expected (H_e) heterozygosities and tests of Hardy-Weinberg equilibrium and linkage disequilibrium were calculated with GenAlEx (Peakall & Smouse, 2006, 2012) and GENEPOP 4.2 (Rousset, 2008). The Benjamini & Hochberg (1995) procedure was used to adjust significant values for multiple comparisons. Tests for the presence of null alleles and scoring errors were performed in MICROCHECKER (Van Oosterhout et al., 2004). GenAlEx was also used to test for the exclusion probabilities for each locus and for increasing combinations of the 17 loci.

RESULTS

For all 17 characterized loci, the repeat motifs matched the probe sequences used, indicating high enrichment efficacy. In the 30 specimens of *Panulirus echinatus* and across the 17 loci, a total of 71 alleles (2-10 alleles per locus), and a mean number of 4.2 alleles per locus, were detected. One of the loci (Pe-L43) presented microvariants (that is, alleles that differed in less than 4 bp). Observed and expected heterozygosities ranged from 0.010 to 0.867 and 0.095 to 0.799, respectively (table I). Mean observed and expected heterozygosities were 0.501 and 0.525, respectively. No significant pairwise linkage disequilibrium was observed among the loci after Benjamini and Hochberg correction (false discovery rate of 0.25), and, with the exception of one locus (Pe-L43), all loci were in Hardy-Weinberg equilibrium. The existence of null alleles in the one locus may account for the Hardy-Weinberg disequilibrium. Null allele frequencies were evaluated by MICROCHECKER to test this hypothesis, which showed the presence of null alleles for locus Pe-L43 (reaching a frequency of 20%).

The probability of excluding two specimens as related when they are not was 90% with the first locus (Pe-L3) and increased to 98.5% with the addition of

the two following loci, Pe-L8 and Pe-L10. A 100% probability of exclusion was attained once 12 of the 17 loci were included in the analysis. Thus, overall, the level of polymorphism and heterozygosity observed in these loci suggests that these markers will be valuable for determining kin relationships.

DISCUSSION

Microsatellite markers are widely used in genetic studies of populations, including many marine organisms (e.g., Milano et al., 2011; Delghandi et al., 2015; López-Márquez et al., 2016; Truelove et al., 2016). In addition, microsatellite development can now be achieved with greater efficiency at lower costs due to recent technological advances and improvements. In this study, using the NGS Illumina MiSeq platform, we have successfully developed 17 loci for *Panulirus echinatus*. All loci were in Hardy-Weinberg equilibrium, except Pe-L43. In this last locus, signature of possible null alleles was found. The presence of null alleles is commonly observed in a variety of abundant marine invertebrate species (Kaukinen et al., 2004) but it is important to analyse and correct it for bias because measures of population genetic structure (F_{ST}) may be overestimated in the presence of null alleles when populations are substantially differentiated (Chapuis & Estoup, 2007). However, in this case, even after accounting for potential heterozygotes with null alleles, the locus still departed from Hardy-Weinberg equilibrium and, thus, other factors might be acting to lead to such disequilibrium (as selection, rate of mutation or reduced population size).

Overall, future studies of the population genetics and structure of this species will greatly benefit from both the number and polymorphism of the new microsatellite markers. Although the long larval period (approximately 6-8 months) of this species potentially facilitates dispersion between Central Atlantic Islands (and along the northeastern coast of Brazil), differentiation among *P. echinatus* populations with a marked genetic structure may still be hypothesized based on the isolation by distance between these islands. This hypothesis, and those about gene flow and connectivity among populations, can now be tested using the markers developed here. Moreover, the highly theoretical probability of connection between islands populations allows for more accurate studies about larval dispersion, retention and recruitment, which may be insightful for improving the management and conservation of this understudied species.

DATA ACCESSIBILITY

Panulirus echinatus contig sequences containing microsatellite loci were deposited into GenBank (MF170655 to MF170671).

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